		$N_1 = W\tan 30 = 15\tan 30 = 8.66 = 8.7$ N (2s.f.)
7	**A**	By conservation of energy, elastic potential energy is converted to kinetic energy and work done against friction. As height is the same at initial and final positions, gravitational potential energy is unchanged. $EPE_{loss} = KE_{gain} + WD_{\text{friction}}$ $\frac{1}{2}kx^2 = \left(KE_{final} - KE_{initial}\right) + fd$ $KE_{final} = \frac{1}{2}kx^2 - fd \qquad \left(\text{since } KE_{initial} = 0\right)$
8	**D**	At different positions along the vertical circular ramp, the normal contact force on the ball